**Practical No 5: Image processing using open CV**

```
!pip install opencv-python numpy matplotlib import
cv2
from google.colab.patches import cv2_imshow
from google.colab import files uploaded =
files.upload() filename =
list(uploaded.keys())[0]

# Step 3: Read image img =
cv2.imread(filename)

# Step 4: Check if loaded if img is None:     print("❌
Error: Image not loaded. Check file format.") else:
    print("☑ Image loaded successfully")
cv2_imshow(img)            print("Original
Image:")     cv2_imshow(img)

    # ----------------------------
    # Step 3: Convert to Grayscale
    # ----------------------------
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
print("Grayscale Image:")     cv2_imshow(gray)

    # ----------------------------
    # Step 4: Gaussian Blur     # ------------
-----------------                          blur      =
```

```
cv2.GaussianBlur(gray,        (5,5),        0)
print("Blurred                          Image:")
cv2_imshow(blur)

    # ----------------------------
    # Step 5: Edge Detection     # -----
------------------------     edges =
cv2.Canny(blur, 100, 200)
print("Edge Detection:")
cv2_imshow(edges)

    # ----------------------------
    # Step 6: Thresholding (Segmentation)
    # ----------------------------
    _, thresh = cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)
print("Threshold Image:")     cv2_imshow(thresh)

    # ----------------------------
# Step 7: Contour Detection
    # ----------------------------
    contours, _ = cv2.findContours(thresh, cv2.RETR_TREE,
cv2.CHAIN_APPROX_SIMPLE)

    contour_img = img.copy()
cv2.drawContours(contour_img, contours, -1, (0,255,0), 2)
print("Contours Detected:")     cv2_imshow(contour_img)
```

Original Image :

Grayscale Image :

Blurred Image :

Edge Detection :

Threshold Image :

Counters Detected :